

MODEL QUESTION PAPER

TED(15)3092

REVISION 2015

THIRD SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

POLYMER SCIENCE

Maximum marks 100

Time 3 hours

Reg No.....

Signature

(PART -A)

I. Answer the following question in one or two sentences

- 1 Give two examples of amorphous and crystalline polymers?
2. Name the instrument used for identification of polymers by physical methods?
3. List the various methods used for determination of molecular weight of polymers?
4. Define the terms Isotactic and syndiotactic polymers with examples of each?
5. Distinguish the terms Blends and Composites? Give one example of each?

(PART -B)

II Answer any FIVE questions each question carries 6 marks

1. Explain TGA ? And also mention its application in polymers ?
2. Explain the different steps involved by chemical method in polymer identification?
- 3 Distinguish between the molecular weights of polymers and simple molecules with examples?
4. How number average molecular weight is determined by Osmometry ?
5. Explain how microstructure of polymers are related to its properties?
6. State the importance of Z-N catalyst in the formation of stereo regularity in polymer formation?
7. Justify that Physical and chemical modifications of polymers are important when considering product manufacturing?

(5x6=30)

PART -C

Answer any one full question from each unit .Each question carry 15 marks

UNIT I

- III a) Compare elastomers, plastics and fibres in terms of Tg and crystallinity (8)
b) Explain the working and application of DSC in polymer identification? (7)
OR
IV a) Write the scheme for final identification of Group VI rubbers? (8)
b) How polymers can be identified by using IR spectroscopy (7)
V a) Explain the determination of MWD by gel permeation chromatography technique (8)
b) Derive an equation for number average molecular weight of polymers (7)

OR

VI a) Distinguish between molecular weight of polymers and that of simple molecules with example(8)

b) Derive an equation for weight average molecular weight of polymers ? (7)

UNIT –III

VII a) Explain tacticity of polymers with example as PP (8)

b) State the importance of Z-N catalyst in stereo regularity of polymers? Give two combinations of Z-N catalyst (7)

OR

VIII a) State the general remarks on polymer microstructure and write cis ,trans.vinyl forms for CR (8) b)Explain the mechanism of formation of stereo regularity in polymers using Z-N catalys ? (7)

UNIT-IV

IX a) Explain the various chemical methods for modifications of polymers (8)

b)Explain polymer reactions such as hydrolysis acidolysis and aminolysis ? (7)

OR

X a)Identify the major differences between reactions of polymers and simple molecules? (8)

b) Explain the modification methods of polymer by blending, grafting and IPN with examples of each (7)

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